# CSE 5717 Project Proposal

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Our project aims to predict the outcomes of English Premier League soccer matches using historical match data. We plan to develop a machine learning model that predicts whether the home team will win, the away team will win, or the match will draw. We plan to use factors such as recent team performance, home-field advantage, goals scored and allowed, and other match statistics to predict future results.

We will use historical English Premier League match data from Football-Data.co.uk. The dataset contains results and statistics from previous Premier League seasons.

We aim to predict matches before they occur, therefore we will be careful to prevent data leakage and only use information that would have been available prior to each match.

We would like our model to predict the outcome of a match with a classification accuracy of around 50%. A uniform random guess would have a 33.3% chance of being correct, and choosing the home team to win would have a 45-46.5% accuracy (homefield advantage). Our goal is for the model to outperform always predicting a home-team win. Based on that, our accuracy target is 50%.

Dataset: https://football-data.co.uk/englandm.php